

Doom PWAD QA Report

Run 16af65cd-4744-4636-9842-8f2f87e1af1c | Map E1M1 | Outcome timeout

Executive Summary

This is an autonomous QA test of E1M1 using an LLM-directed test harness. The test was designed to evaluate map navigation and functional gameplay, but the run resulted in a timeout after 11 seconds due to a lack of spawnable content at the selected difficulty level.

Pass/fail:

map_navigation: PASS

combat_engagement: FAIL

resource_balance: PASS

secret_coverage: PASS

overall_verdict: PARTIAL

- The automated playthrough moved between 3 distinct position clusters, satisfying the navigation criteria.
- The playthrough failed combat engagement because spawned_enemy_count was 0 while the raw_enemy_count indicated enemies should exist.
- Resource balance passes automatically as spawned_enemy_count is 0.
- Secret coverage passes because there are 0 secret sectors in the map.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated map E1M1 using the Freedoom1 IWAD at difficulty level 5. Static analysis identified 8 enemies (Imps, Zombiemen, Shotgun Guys) and 1 item (Green Armor), with a map dimension of 2528x1056 units across 5 sectors. There are no secret sectors present in the map geometry.

The test environment utilized ViZDoom as the game engine, with the gemini-3.1-flash-lite LLM model coordinating actions through the MCP interface. The run was conducted at difficulty level 5 with a max_ticks cap of 260.

Metric	Value	Metric	Value
Status	completed	Duration	11
Actions	2	LLM calls	2
Final HP	100	Kills	0
Position samples	3	Movement units	1450.6
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/16af65cd-4744-4636-9842-8f2f87e1af1c.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 5	Notes
Enemies	8	0	8 hidden by skill/multiplayer flags
Items	1	0	1 hidden by skill/multiplayer flags
Secrets	0	0	Sector special 9
Geometry	35 linedefs	5 sectors	2528 x 1056 units

Selected-difficulty estimate: easy. Health ratio: 0.0. Ammo ratio: 0.0.

Coverage Evaluation

The coverage was limited to navigation; while the automated playthrough moved across three distinct position clusters, it did not encounter any entities. Gameplay coverage is insufficient to verify combat or resource balance mechanics. The static analysis indicates the map contains content that was inaccessible due to difficulty flag configuration.

Objectives Covered

- **Navigate the entire map:**

Objectives Omitted

- **Engage all enemy types:** 0 enemies spawned due to difficulty 5 configuration.
- **Document resource balance issues:** 0 items spawned due to difficulty 5 configuration.

Uncovered attributes: Combat mechanics, pickup interactions, and resource management systems were not tested as the map was effectively empty at the chosen difficulty setting.

Defects And Risks

A total of 1 defect was detected, categorized as Severity 2. The most impactful issue is the difficulty-spawn mismatch where all content is hidden at the test difficulty level.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
2	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 5, 8 of 8 enemies and 1 of 1 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.

Risk Areas

- **Global spawn flags:** medium Map content is completely inaccessible at difficulty 5, potentially hindering intended gameplay experiences.

Activity Log Summary

1) Static analysis performed, 2) Map exploration initiated, 3) Episode concluded (ticks 17-225).
Combat phase was absent because the map contained no spawned enemies at the tested difficulty.

Limitations And Recommendations

No specific interactions with map triggers or entities were tested due to the absence of spawned content.

Good Quality Areas

- **Navigation pathing:** The automated playthrough successfully moved through the map space without encountering geometry-based softlocks.